

Enabling High-Goodput LLM Inference Serving with Heterogeneous SLO-Aware Speculative Decoding

Paper #1664

Background and Motivation

Modern *large language model* (LLM) [17, 18, 25] serving is becoming shared infrastructure for applications with different latency requirements [5]. Interactive dialogue [14, 19] requires a steady token stream, code completion [4, 13] demands rapid generation for seamless workflows, while document processing [1, 20] and complex reasoning [22, 23, 29] can tolerate more relaxed generation rates, prioritizing quality over immediacy. The increasing multiplexing of these workloads on shared GPU clusters demands that LLM serving systems support heterogeneous *service-level objectives* (SLOs) [6, 15, 28], rather than imposing a uniform generation target.

For streaming LLM services, responsiveness after the first token is largely governed by *time per output token* (TPOT), which captures the sustained decoding rate of a request. Heterogeneous TPOT SLOs shift the serving objective from maximizing throughput to allocating decoding resources under SLO constraints. Continuous batching [8, 24] remains essential for improving GPU utilization, but conventional batching policies typically advance all requests uniformly at each decoding iteration, providing limited control over each request’s progress toward its individual TPOT target, as illustrated in Figure 1(a). When requests with tight and relaxed TPOT SLOs are batched together, such uniform progress can overallocate decoding work to requests with ample latency slack, while requests close to SLO violations receive insufficient priority. As a result, throughput no longer reflects the effective serving capacity under heterogeneous SLOs. The optimization objective should instead be **SLO-constrained goodput** [28], which counts only the tokens generated by requests that meet their assigned TPOT SLOs [26].

Speculative decoding [2, 3, 9, 12, 21] provides a promising mechanism for adaptive progress control under heterogeneous SLOs. It uses a lightweight draft model to propose multiple draft tokens and then verifies them with the target model in one iteration, allowing each request to yield a variable number of accepted tokens per decoding step, as illustrated in Figure 1(b). For a single request, its sustained generation latency can be approximated as $\text{TPOT} \approx \frac{T_{\text{draft}} + T_{\text{verify}}}{\mathbb{E}[A]}$, where T_{draft} and T_{verify} are the draft and verification latencies, and $\mathbb{E}[A]$ is the expected number of accepted tokens per speculative iteration. This relation reveals a tunable tradeoff governed by the *speculative budget*, i.e., the number of draft tokens proposed and verified in each speculative iteration. A larger budget can improve $\mathbb{E}[A]$, but also increases draft and

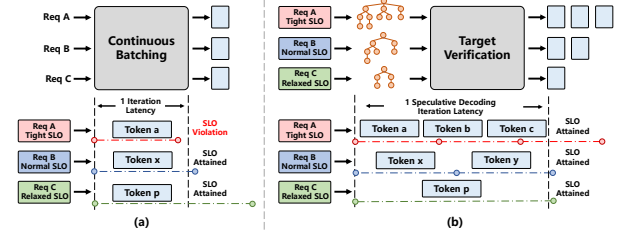


Figure 1. Motivation for SLO-aware speculative serving.

verification latency. This tradeoff naturally supports SLO-aware control, where requests with tight TPOT SLOs receive larger budgets to accelerate progress, while requests with relaxed SLOs receive smaller budgets to conserve computation.

Limitations of Current Approaches

Existing SLO-aware speculative serving systems [7, 10] exploit this tradeoff by assigning different budgets to requests with different latency targets. However, they retain a serial draft-then-verify pattern where the next draft stage must wait for verification to finish. Consequently, aggressive budgets directly increase the draft latency exposed in each iteration. This serial coupling becomes more severe in batched serving, where concurrent requests share compute resources and progress under synchronization boundaries. If the total batch-level speculative capacity is fixed to bound iteration latency, each request receives a smaller effective budget as the batch grows, reducing accepted progress per round. If each request keeps its own budget, the total number of draft and verification candidates grows with batch size, substantially increasing iteration latency for all requests. Our experiments confirm this latency-progress dilemma: as the active batch size grows from 8 to 64, the mean number of accepted tokens per request drops from 4.1 to 1.7.

Recent asynchronous parallel speculative decoding systems [11, 27] attempt to hide draft latency by overlapping drafting with target verification. However, this hidden drafting window is susceptible to prefix invalidation because draft tokens generated from an unconfirmed prefix are useful only if the target model fully accepts that prefix. Otherwise, the continuation is discarded. In our PEARL evaluation with draft length 4, only 32.4% of hidden draft tokens are accepted, while 38.4% are invalidated by prefix rejection and 29.2% are rejected during verification. More importantly, existing parallel mechanisms optimize overlap efficiency rather than allocate the hidden drafting window based on TPOT urgency. As batch size grows, tight-SLO attainment drops sharply while relaxed requests remain mostly satisfied. This

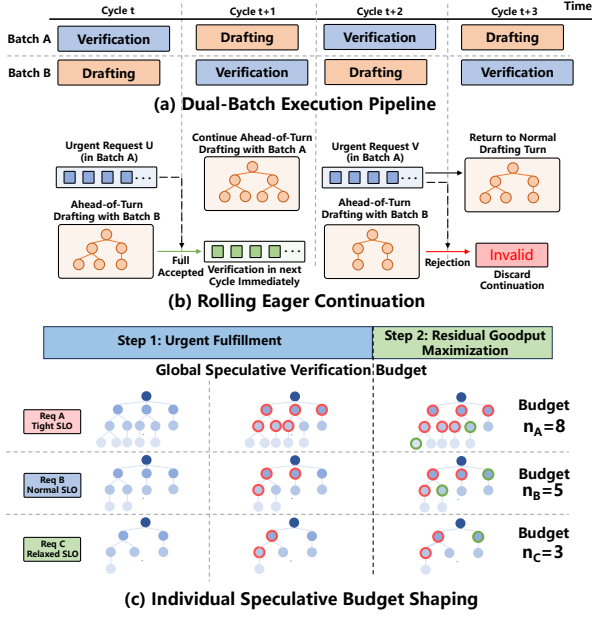


Figure 2. Overview of SpecRhythm.

indicates that the hidden drafting capacity is not directed to the requests that most need timely progress.

Key Ideas and Design

We present **SpecRhythm**, a speculative LLM serving system designed to maximize goodput. The key idea is to separate the execution structure that creates speculative capacity from the control policy that allocates that capacity. At the execution layer, SpecRhythm builds a stable batch-level overlap structure and treats the resulting drafting windows as allocatable resources. At the control layer, it decides which requests should draft ahead of their normal batch turn and how much speculative budget each request should receive.

SpecRhythm realizes this design through three integrated techniques. First, the **Dual-Batch Execution Pipeline** organizes active requests into two alternating logical batches. In each service step, the target model verifies the draft tokens of one batch, while the draft model generates new draft tokens for the other batch. In the next step, the two batches swap roles. This structure hides drafting work within the complementary verification window and provides a predictable batch-level drafting window for scheduling control.

Second, **Rolling Eager Continuation** exploits the available overlap window to advance urgent requests before their scheduled batch turn. Standard dual-batch rotation forces requests to wait until their assigned drafting stage, which lacks the flexibility needed for those near their SLO boundary. SpecRhythm evaluates progress gaps right before target verification. For highly urgent requests, it initiates ahead-of-turn drafting by injecting them into the active drafting window of the alternating batch. If the current prefix is accepted, these eagerly drafted tokens immediately serve as ready candidates; otherwise, they are discarded to preserve

correctness. By hiding draft latency, this technique accelerates token generation for latency-critical requests.

Third, **Individual Speculative Budget Shaping** determines how candidate tokens are distributed under a bounded verification budget. Target verification becomes more expensive as the number of candidate tokens grows, so SpecRhythm uses offline profiling to identify a roofline-guided verification budget that keeps verification within a low-latency region. Within this budget, SpecRhythm first allocates candidates to urgent requests with projected progress gaps, prioritizing high-confidence tokens that are more likely to be accepted. Any remaining candidate slots are then assigned to high-confidence candidates across requests, weighted by residual urgency. This two-step policy avoids wasting speculative capacity on requests with sufficient slack or low expected acceptance, while still improving overall goodput.

Experimental Results

We implement SpecRhythm on top of nano-vLLM and evaluate it on a single node with 4× NVIDIA A100 80GB GPUs, pairing the Llama3.1-70B-Instruct and Qwen3-32B target models with the Llama3.2-1B-Instruct and Qwen3-0.6B draft models, respectively. Baselines include SpecInfer (serial speculative decoding) [12], AdaServe (SLO-aware serial speculative decoding) [10], and PEARL (asynchronous parallel speculative decoding) [11]. The default workload mixes tight (code completion [4]), normal (chatbot [16]), and relaxed (summarization [1]) requests at a 6:2:2 ratio.

Across diverse workloads, SpecRhythm consistently improves goodput and SLO attainment. At an active batch size of 64, it delivers 721.3 tokens/s with 80.7% attainment on Llama, and 764.8 tokens/s with 76.4% attainment on Qwen3, outperforming AdaServe by up to 1.27× in goodput and 1.31× in attainment. Under dynamic online traffic at 4 RPS, SpecRhythm sustains 603.3 tokens/s on Llama and 711.0 tokens/s on Qwen3 while maintaining the highest SLO attainment among all evaluated baselines. Similar gains hold under stricter SLO targets and higher fractions of tight requests, demonstrating that SpecRhythm remains robust as heterogeneous SLO pressure increases. An ablation study further demonstrates that overlapping execution alone is insufficient. Specifically, deploying only dual-batching causes goodput to drop from 182.4 to 150.4 tokens/s. However, incorporating rolling continuation successfully reclaims this loss and boosts goodput to 392.5 tokens/s, while further integrating budget shaping elevates the performance to a peak of 723.0 tokens/s.

Summary

Overall, SpecRhythm advances heterogeneous LLM serving with a batch-level SLO-aware speculative decoding design. A dual-batch execution pipeline creates predictable drafting capacity, while rolling eager continuation and individual budget shaping allocate it by runtime urgency and draft confidence to improve goodput and SLO attainment.

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